

LAB PROGRAMS

QUEUE

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS ENGINEERING logo, the text 'SIMATS Online Programming Lab', and 'DS PROGRAMMING'. A user profile for 'Nithiyasri' is visible in the top right. The left sidebar contains a 'QUESTIONS' section with '10 / 10 completed' and a dropdown menu set to 'Queue Using Array'. Below this is the title 'QUEUE USING ARRAYS' and a description: 'Write a C program to implement a Queue using Arrays with operations such as insertion (enqueue), deletion (dequeue), display, and peek. The program should simulate a ticket booking counter where customers arrive in sequence and are served on a first-come first-served basis.' The main 'PROGRAM EDITOR' area shows a C program for a queue implemented using an array. The code includes headers, defines MAX as 100, declares an array and front/rear pointers, and implements enqueue and dequeue functions. The 'OUTPUT' section at the bottom displays 'Queue Elements: 101 102 103'. The right sidebar has an 'INPUT' section with the text 'Your INPUT goes here!'.

QUESTIONS 10 / 10 completed

Queue Using Array

QUEUE USING ARRAYS

Write a C program to implement a Queue using Arrays with operations such as insertion (enqueue), deletion (dequeue), display, and peek. The program should simulate a ticket booking counter where customers arrive in sequence and are served on a first-come first-served basis.

PROGRAM EDITOR

```
1 #include <stdio.h>
2 #define MAX 100
3
4 int queue[MAX];
5 int front = -1;
6 int rear = -1;
7
8 // Function to add a customer (enqueue)
9 void enqueue(int value) {
10     if (rear == MAX - 1) {
11         printf("Queue Overflow\n");
12     } else {
13         if (front == -1) {
14             front = 0;
15         }
16         rear++;
17         queue[rear] = value;
18     }
19 }
20
21 // Function to remove a customer (dequeue)
```

OUTPUT

Queue Elements: 101 102 103

INPUT

Your INPUT goes here!

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QUESTIONS 10 / 10 completed

Queue Using Linke

QUEUE USING LINKED LISTS

Write a C program to implement a Queue using Linked Lists for managing patients waiting in a hospital consultation room. The program should support enqueue, dequeue, display, and front operations without any fixed size limitation. Sample Input: Insert Patient ID: 201 202 203

PROGRAM EDITOR

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 // Structure for a patient node
5 struct Node {
6     int patientID;
7     struct Node* next;
8 };
9
10 // Structure for the queue
11 struct Queue {
12     struct Node *front, *rear;
13 };
14
15 // Function to create a new node
16 struct Node* newNode(int k) {
17     struct Node* temp = (struct Node*)malloc(sizeof(struct Node));
18     temp->patientID = k;
19     temp->next = NULL;
20     return temp;
21 }
```

OUTPUT

Queue Elements: 201 202 203

INPUT

Your INPUT goes here!

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QUESTIONS 10 / 10 completed

Double Ended Queue ▾

DOUBLE ENDED QUEUE

Write a C program to implement a Double Ended Queue (Deque) using Arrays where insertion and deletion can be performed from both front and rear ends. Consider a warehouse inventory management system where goods can be added or removed from either side based on priority. Sample

PROGRAM EDITOR

C ▾

Run Save

```
1 #include <stdio.h>
2 #define MAX 10
3
4 int deque[MAX];
5 int front = -1;
6 int rear = -1;
7
8 // Function to check if the deque is full
9 int isFull() {
10     return ((front == 0 && rear == MAX - 1) || (front == rear + 1));
11 }
12
13 // Function to check if the deque is empty
14 int isEmpty() {
15     return (front == -1);
16 }
17
18 // Insert at Front
19 void insertFront(int val) {
20     if (isFull()) {
21         printf("Queue Overflow\n");
```

OUTPUT

Deleted Element: 200

INPUT

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QUESTIONS 10 / 10 completed

Priority Queue Usir ▾

PRIORITY QUEUE USING ARRAYS

Write a C program to implement a Priority Queue using Arrays for processing emergency service requests. Requests with higher priority must be processed before lower-priority requests irrespective of arrival order. Sample Input: Insert (101, Priority 3), Insert (102, Priority 4), Insert

PROGRAM EDITOR

C ▾

Run Save

```
1 #include <stdio.h>
2 #define MAX 100
3
4 typedef struct {
5     int id;
6     int priority;
7 } Request;
8
9 Request pq[MAX];
10 int size = 0;
11
12 void insert(int id, int priority) {
13     if (size >= MAX) {
14         printf("Queue Overflow!\n");
15         return;
16     }
17
18     pq[size].id = id;
19     pq[size].priority = priority;
20     size++;
21 }
```

OUTPUT

Processed Request: 102

INPUT

INPUT



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QUESTIONS 10 / 10 completed

Queue Data Struct

QUEUE DATA STRUCTURE

Write a C program to simulate a Railway Reservation Waiting List using Queue data structure. Passengers enter the waiting queue and are allocated seats as cancellations occur. The program should maintain passenger details and process requests in FIFO order.
Compile Insert Add

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 #define MAX 5
6
7 struct Passenger {
8     int id;
9 };
10
11 struct Queue {
12     struct Passenger items[MAX];
13     int front;
14     int rear;
15 };
16
17 void initQueue(struct Queue *q) {
18     q->front = -1;
19     q->rear = -1;
20 }
21
```

OUTPUT

Error: Execution timeout (>5 seconds) - possible infinite loop

INPUT

Your INPUT goes here!



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QUESTIONS 10 / 10 completed

Queue

QUEUE

Write a C program to implement a Queue for managing online food delivery orders where orders are received and processed in the order they arrive. The program should allow adding new orders, processing completed orders, and displaying pending orders.
Compile Insert Add

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 int isEmpty() {
9     return front == -1 && rear == -1;
10 }
11
12 int isFull() {
13     return rear == MAX - 1;
14 }
15
16 void enqueue(int value) {
17     if (isFull()) {
18         printf("Queue Overflow! Cannot add more orders.\n");
19         return;
20     }
21
```

OUTPUT

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QUESTIONS 10 / 10 completed

Queue Using Linke

QUEUE USING LINKED LISTS -2

Write a C program to implement a Queue using Linked Lists for managing network packets in a router. Packets should be transmitted in the order received, and the program must support dynamic packet insertion and removal. Sample Input: Insert Packets 11 12 13 Transmit

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 // Structure to represent a network packet
5 struct Packet {
6     int packetID;
7     struct Packet* next;
8 };
9
10 // Pointers to manage the queue
11 struct Packet* front = NULL;
12 struct Packet* rear = NULL;
13
14 // Function to insert a packet (Enqueue)
15 void insertPacket(int id) {
16     struct Packet* newNode = (struct Packet*)malloc(sizeof(struct Packet));
17     newNode->packetID = id;
18     newNode->next = NULL;
19
20     if (rear == NULL) {
21         front = rear = newNode;
```

OUTPUT

Transmitted Packet: 11

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Nithya 19257

QUESTIONS 10 / 10 completed

Circular Queue

CIRCULAR QUEUE

Write a C program to implement a Circular Queue for a traffic signal management system where vehicles are processed lane by lane in a cyclic manner. The system should continuously rotate through available lanes while maintaining queue order. Sample Input: Vehicles 21 22 23

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define SIZE 5
5
6 int items[SIZE];
7 int front = -1, rear = -1;
8
9 // Check if the queue is full
10 int isFull() {
11     if ((front == rear + 1) || (front == 0 && rear == SIZE - 1)) return 1;
12     return 0;
13 }
14
15 // Check if the queue is empty
16 int isEmpty() {
17     if (front == -1) return 1;
18     return 0;
19 }
20
21 // Add an element
```

OUTPUT

Vehicle Arrived: 21

INPUT

Your INPUT go here!

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QUESTIONS 10 / 10 completed

Queue-2

QUEUE-2

Write a C program to implement a Queue for managing examination answer sheet evaluation. Answer sheets submitted by students are evaluated in the same sequence in which they are received. Sample Input: Sheet IDs 401, 402, 403 submitted, evaluate ans sheet

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 100
5
6 struct Queue {
7     int items[MAX];
8     int front;
9     int rear;
10 };
11
12 // Initialize the queue
13 void initQueue(struct Queue* q) {
14     q->front = -1;
15     q->rear = -1;
16 }
17
18 // Check if the queue is full
19 int isFull(struct Queue* q) {
20     return q->rear == MAX - 1;
21 }
```

OUTPUT

INPUT

Your INPUT goes here!

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QUESTIONS 10 / 10 completed

Queue Using Linke

QUEUE USING LINKED LIST-3

Write a C program to simulate a supermarket billing queue using Linked List implementation. Customers join the billing queue and are billed in FIFO order. The queue size should dynamically increase as customers arrive. Sample Input: Customers 11, 12, 13 enter, ans customer

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 // Structure to represent a customer node in the linked list
5 struct Node {
6     int customerID;
7     struct Node* next;
8 };
9
10 // Structure to represent the queue
11 struct Queue {
12     struct Node *front, *rear;
13 };
14
15 // Function to create a new node
16 struct Node* newNode(int k) {
17     struct Node* temp = (struct Node*)malloc(sizeof(struct Node));
18     temp->customerID = k;
19     temp->next = NULL;
20     return temp;
21 }
```

OUTPUT

Billed Customer: 11

INPUT

Your IN here!